

Environments: Choose the right room to accomplish a task

More often than not, to accomplish a task you need several tools (i.e. software), and according to the task itself, you need specific tools.

Solving different tasks will lead you to continuously add new software that increase the probability of software conflicts.

Creating an environment for each task (or small group of heterogeneous tasks) will help to avoid software conflicts.

conda is the tool to easily manage environments.

Installing conda

First of all you need to install the package manager. I suggest to use miniconda3.

To install it, follow the instructions at the following link [Miniconda — conda documentation](#)

Creating an environment

To create a new environment is simple as this:

```
conda create -n MyEnvironmentName
```

then you need to activate it

```
conda activate MyEnvironmentName
```

and populate it with the available software of your choice

```
conda install python=3.11 pandas matplotlib
```

With the line above, you will install python (version 3.11), its two libraries matplotlib and pandas, with all the necessary programs (dependencies).

From now on, whenever you need to use python together with pandas or matplotlib, it is enough to activate the environment and use them.

Remember that the programs installed within an environment are available only within that environment.

Using an environment

To “enter” an environment you need to explicitly activate it:

```
conda activate environmentName
```

Once finished, you can “exit” the environment by deactivating it:

```
conda deactivate
```

To obtain the list of all the available environments, digit

```
conda info --envs
```

Sharing an environment

It is possible to save a snapshot of an environment in order to share it, or to port it on another computer.

```
conda activate MyEnvironmentName  
conda env export > environment.yml
```

then, you can create an exact copy of it running

```
conda env create -f environment.yml
```

Removing an environment

To completely remove an environment from your computer, digit

```
conda env remove --name MyEnvironmentName
```

Conda channels

A conda channel is a compilation of packages. Some of them are thematic bioconda or tries to have the most up to date cross platform version of the packages conda-forge

To add a channel to an environment run

```
conda config --append channels conda-forge
```

Example:

Create an environment to perform exploratory data analyses with python. Let's say we need python (3.11), ipython (to ease the use of python), pandas (to mangle the data), openpyxl (to write *.xlsx) and matplotlib to visualize the information.

```
conda create -n dataAn python=3.11
```

```
conda activate dataAn
```

```
conda install ipython pandas openpyxl matplotlib
```

or from the file condaEnvDataAnalysisMinimal.yml

```
conda create -f condaEnvDataAnalysisMinimal.yml
```